

# Sofia ML Regression 2026 Spring

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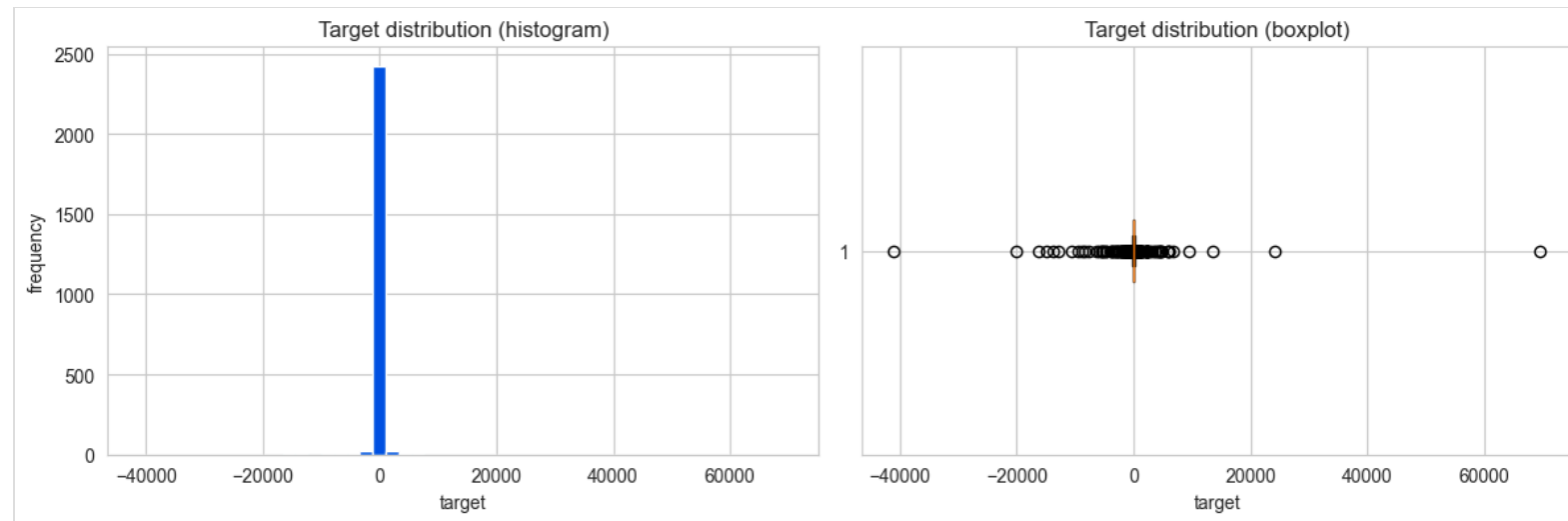
Kaggle: <https://www.kaggle.com/baek007>

## TL;DR

- **Problem:** 15 anonymized features → one real-valued target, scored by  $R^2$ . Very weak signal, extreme outliers.
- **Main idea:** don't fight outliers with a fancier model — clean the *training* target (winsorize) and keep a simple Ridge.
- **Result:** Public LB 0.04843, beat the baseline (0.04599); was #1 on public for a while.
- **Final twist:** on the private split my rank didn't move at all (public 6th → private 6th) while the top reshuffled wildly.

# EDA

- 2500 train rows × 15 anonymized features (x0–x14), ~1.2% missing per column.
- The target is brutal:
  - range -41,008 to +69,628, but most values sit in -50...+50
  - skewness 13.6, kurtosis 708
  - 0.4% of rows have  $|y| > 10,000$
- Weak signal: max |correlation| with the target is 0.25 (x9). Features barely correlate with each other either.



*Target distribution — extreme tails*

**Takeaway:** weak signal + strong outliers → a linear model gets wrecked by a handful of extreme rows.

# Experiments & what failed (and why)

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I started inside the lecture toolkit (Lec 02/03/06/07) and let the failures guide me.

- **Plain Linear Regression** →  $CV R^2 = -0.46$ . One fold hit  $-2.41$  because the outliers landed there — the "sensitive to outliers" weakness from Lec 07.
- **Polynomial + Ridge** →  $-0.84$ . The polynomial terms just amplified the outliers.
- **Log-transformed target** →  $-26$ . The  $\exp()$  inverse blew small errors up — one test prediction reached  $-309,744$ .
- **Swept Ridge / LASSO / ElasticNet / k-NN** → they all converged around **+0.021**.

**Motivation that carried the project:** when 3 different model families agree on the same ceiling, the limit is the *data*, not the model. So I stopped trying to be clever with the model.

## The fix: winsorize the target

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- **Insight:** outliers in the training data corrupt the fit, and I can't predict them in the test set anyway. So remove their influence from *training only* — never touch the test data.
- **Method:** clip the training target to the **[0.5%, 99.5%] percentiles**, then fit **Ridge ( $\alpha=1$ )**.
- **Clip level = bias–variance tradeoff:** too hard = lose information, too light = the outliers come back. [0.5, 99.5] was the sweet spot.
- **Evaluation note:** I trusted **out-of-fold (OOF)  $R^2$** , not `cross_val_score`, because one bad fold tanks the averaged version.

# Results + the big lesson

**Final model:** median impute → standardize → winsorize[0.5, 99.5] → Ridge( $\alpha=1$ )

**Calibration (linear family):** OOF +  $\sim 0.018 \approx$  Public LB

| Version     | Approach                                             | OOF            | Public         |
|-------------|------------------------------------------------------|----------------|----------------|
| v04         | ElasticNet (no winsor)                               | +0.0140        | +0.0320        |
| v07         | Winsor[1–99] Ridge( $\alpha 10$ )                    | +0.0229        | +0.0434        |
| <b>v07b</b> | <b>Winsor[0.5–99.5] Ridge(<math>\alpha 1</math>)</b> | <b>+0.0327</b> | <b>+0.0484</b> |
| v07d        | Winsor[0.1–99.9] Ridge( $\alpha 0.01$ )              | +0.0393        | +0.0190        |

v07d: OOF went UP but LB crashed → OOF can lie on this data.

**Private shake-up (final)**

| Team             | Public        | Private       | Move              |
|------------------|---------------|---------------|-------------------|
| Pawel (pub #1)   | 0.0560        | 0.0538        | ↓ 5th             |
| yifeili00        | −0.093        | 0.0625        | ↑ 3rd             |
| JixiangDing      | −0.093        | 0.0624        | ↑ 4th             |
| <b>Me (v07b)</b> | <b>0.0484</b> | <b>0.0501</b> | <b>0 — stayed</b> |

Two teams negative on public ended up 3rd/4th on private. I was the only top entry whose rank didn't move.

**Lesson:** when the signal is this weak, OOF lies *and* the public LB lies. The real signal is **stable generalization**, not a high score on one split.

# Reflection + code

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## What worked

- Honest EDA → understood the outlier problem first
- Multiple model families to find the true ceiling
- Winsorizing — fix the data, not the model
- OOF  $R^2$  over `cross_val_score`

## What didn't (all documented)

- Polynomial features, log-target transform, heavy feature engineering, too-light winsor, trees/kernel methods

## On the dataset (constructive)

- $R^2$  ceiling  $\sim 0.05$ – $0.08$  → 90%+ of the variance is irreducible noise. Train/test same distribution (adversarial AUC  $\approx 0.49$ ) — not overfitting, the signal is just weak.
- That's why the final ranking shuffled so much: rank was driven by variance as much as skill.
- **Honest note:** I beat the baseline on public (0.0484 vs 0.0460) but landed just under it on private (0.0501 vs 0.0526). Most teams above the private baseline jumped from the bottom of the public board — variance, not a stable model. Mine was the only top entry that didn't move on either split.
- A slightly stronger signal would let the board reward modeling over luck.

Code: <https://github.com/ethanbaek0112/sofia-ml-regression> — `src/phase7b_winsor_all.py` reproduces the final submission.